

Visa Application Approval Prediction

EasyVisa Project: Ensemble Techniques
and Model Tuning

Business Presentation

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Executive Summary

The profile of the applicants for whom the visa status can be approved:

Primary information to look at:

- Education level - At least has a Bachelor's degree - Master's and doctorate are preferred.
- Job Experience - Should have some job experience.
- Prevailing wage - The median prevailing wage of the employees for whom the visa got certified is around 72k.

Secondary information to look at:

- Unit of Wage - Applicants having a yearly unit of wage.
- Continent - Ideally the nationality and ethnicity of an applicant shouldn't matter to work in a country but previously it has been observed that applicants from Europe, Africa, and Asia have higher chances of visa certification.
- Region of employment - Analysis shows that Midwest region have more chances of visa approval.

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Executive Summary

The profile of the applicants for whom the visa status can be denied:

Primary information to look at:

- Education level - Doesn't have any degree and has completed high school.
- Job Experience - Doesn't have any job experience.
- Prevailing wage - The median prevailing wage of the employees for whom the visa got certified is around 65k.

Secondary information to look at:

- Unit of Wage - Applicants having an hourly unit of wage.
- Continent - Ideally the nationality and ethnicity of an applicant shouldn't matter to work in a country but previously it has been observed that applicants from South America, North America, and Oceania have higher chances of visa applications getting denied.

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Executive Summary

- Additional information of employers and employees can be collected to gain better insights. Information such as:
 - Employers: Information about the wage they are offering to the applicant, Sector in which company operates in, etc
 - Employee's: Specialization in their educational degree, Number of years of experience, etc

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Business Problem Overview and Solution Approach

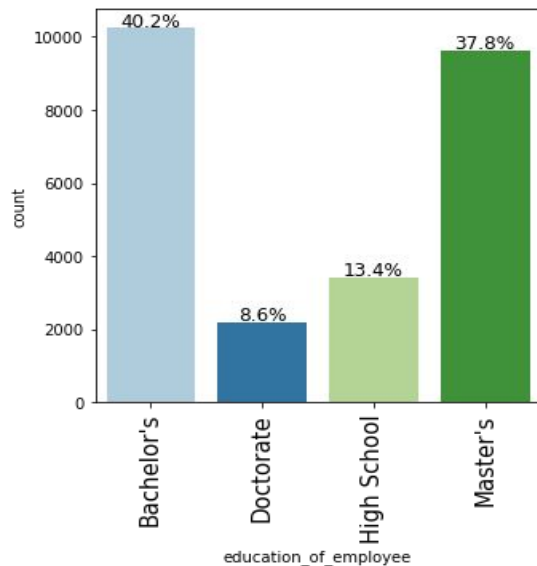
- OFLC processes job certification applications for employers seeking to bring foreign workers into the United States and grants certifications in those cases where employers can demonstrate that there are not sufficient US workers available to perform the work at wages that meet or exceed the wage paid for the occupation in the area of intended employment.
- The process of reviewing every case is becoming a tedious task as the number of applicants is increasing every year. In FY 2016, the OFLC processed 775,979 employer applications for 1,699,957 positions for temporary and permanent labor certifications which was a nine percent increase in the overall number of processed applications from the previous year.
- The task at hand to analyze the data provided to facilitate the process of visa approvals by building a machine learning model and to recommend a suitable profile for the applicants for whom the visa should be certified or denied based on the drivers that significantly influence the case status.

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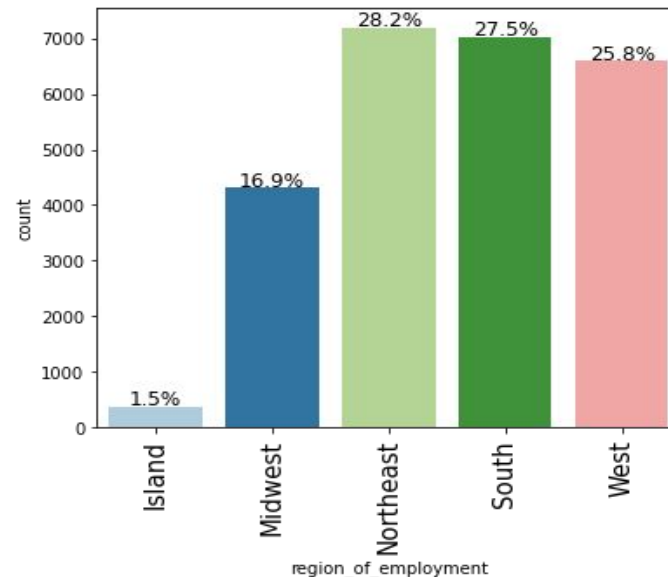
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Exploratory Data Analysis



- 40.2% of the applicants have a bachelor's degree, followed by 37.8% having a master's degree.
- 8.6% of the applicants have a doctorate degree.



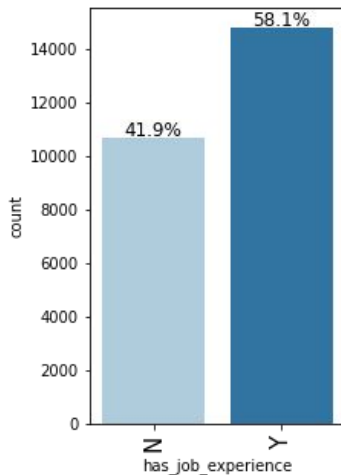
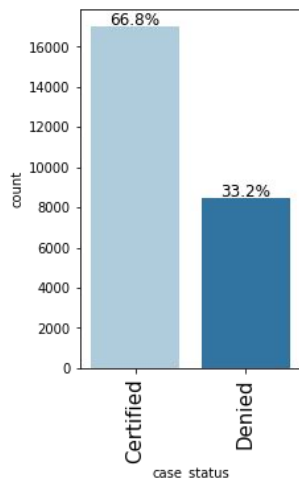
- Northeast, South, and West have almost equal percentages of applicants.
- The Island regions have only 1.5% of the applicants.

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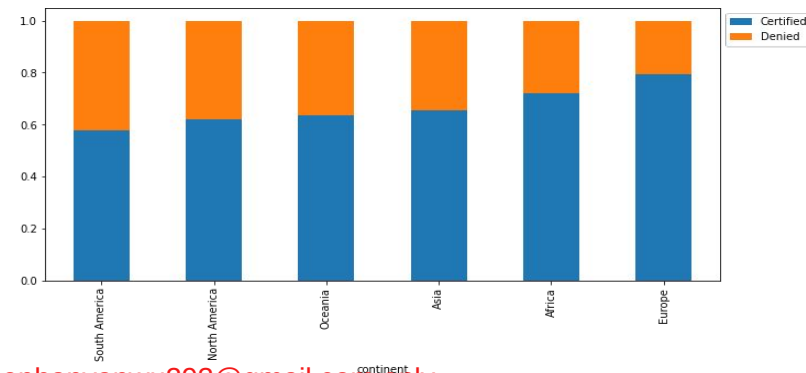
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Exploratory Data Analysis



- Applications from **Europe** and **Africa** have a higher chance of getting certified.
- Around 80% of the applications from **Europe** are certified.
- **Asia** has the third-highest percentage (Around 60%) of visa certification and has the highest number of applications.

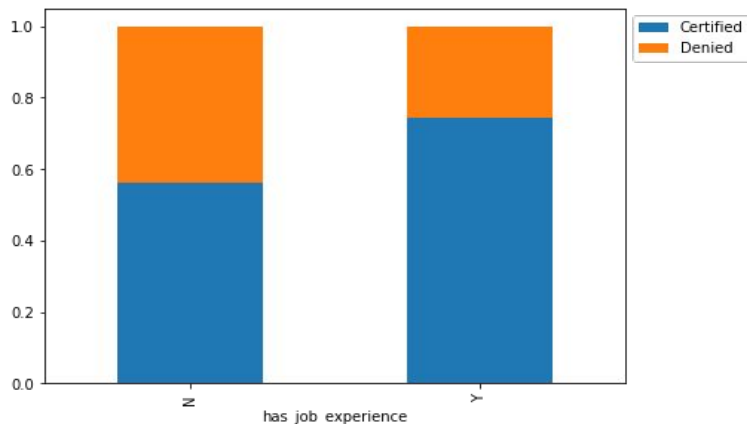
- 66.8% of the visas were certified
- 58.1% of the applicants have job experience.



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Exploratory Data Analysis



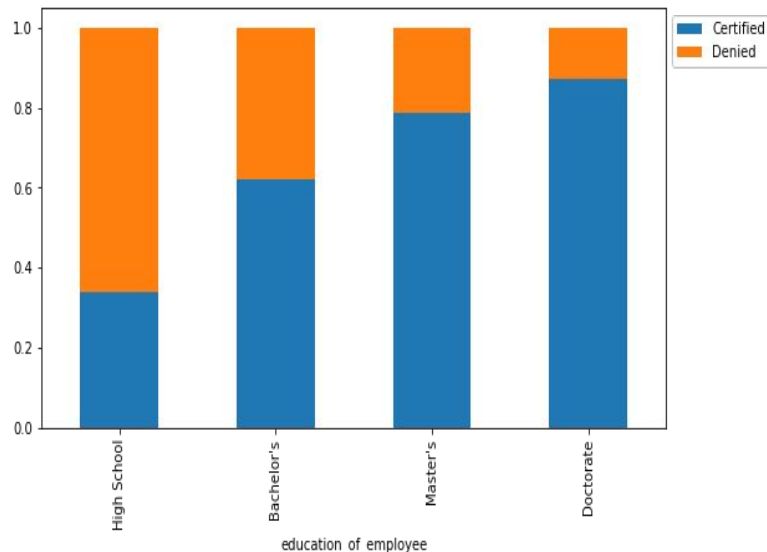
- **Having job experience** seems to be a key differentiator between visa applications getting certified or denied.
- Around 80% of the applications were certified for the applicants who have some job experience as compared to the applicants who do not have any job experience.
- Applicants without job experiences saw only 60% of the visa applications getting certified.

- **Unit of prevailing wage** is an important factor for differentiating between a certified and a denied visa application.
- If the unit of prevailing wage is Yearly, there's a high chance of the application getting certified.
- Around 75% of the applications were certified for the applicants who have a yearly unit of wage. While only 35% of the applications were certified for applicants who have an hourly unit of wage.



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Exploratory Data Analysis



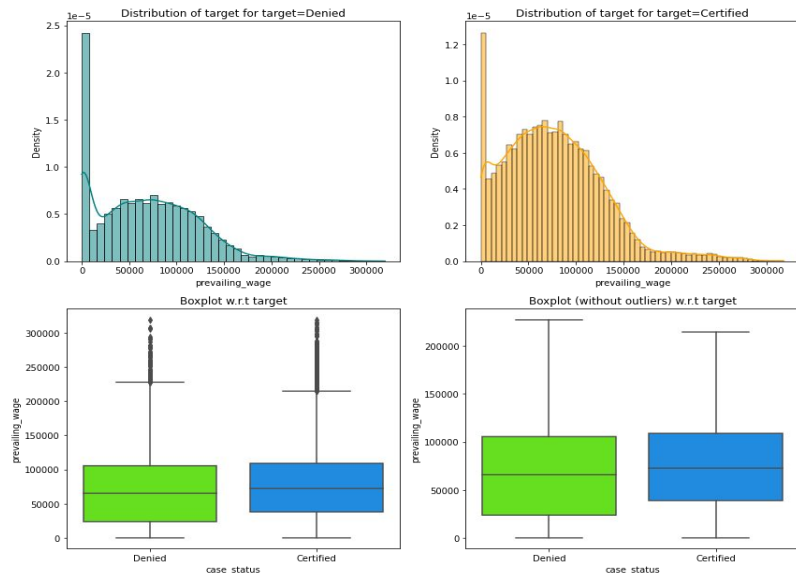
- **Education** seems to have a positive relationship with the certification of visa that is higher the education higher are the chances of visa getting certified.
- Around 85% of the visa applications got certified for the applicants with Doctorate degree. While 80% of the visa applications got certified for the applicants with Master's degree.
- Around 60% of the visa applications got certified for applicants with Bachelor's degrees.
- Applicants who do not have a degree and have graduated from high school are more likely to have their applications denied.

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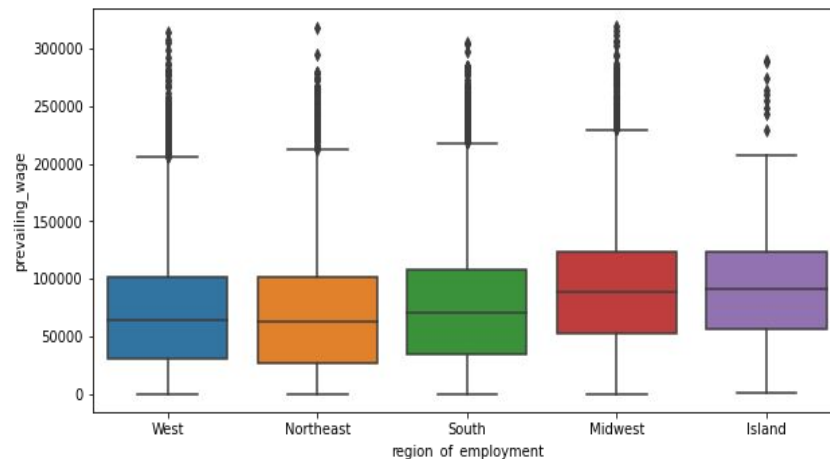
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Exploratory Data Analysis



- The median prevailing wage for the **certified** applications is slightly higher as compared to **denied** applications.



- **Midwest** and **Island** regions have slightly higher prevailing wages as compared to other regions.
- The distribution of prevailing wage is similar across **West**, **Northeast**, and **South** regions.

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Data Preprocessing

- There were no null and duplicate values in the dataset.
- The case_id column was dropped from the dataset since all the values in that column were unique.
- There were some negative values in the number of employees column which were replaced by their absolute values.
- There were quite a few outliers in the data, however, they were not treated since they were genuine values.
- The values in case_status column were encoded - if the application is certified, the value was changed to 1. Similarly, if the application was denied, the value was changed to 0.
- For preparing the data for modeling, we first split the data into 2 parts, temporary and test in an 80:20 ratio as an intermediary step. After this, the temporary dataset is again split into training and validation sets in the ratio of 75:25.

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Model Evaluation Criterion

- We are interested in predicting whether a visa application is certified or denied
- The two cases here where the model can make wrong predictions and the corresponding consequences are:
 - **False Positives:** If a visa is certified when it has to be denied, the wrong employee will get the job position while US citizens will miss the opportunity to work in that position.
 - **False Negatives:** If a visa is denied when it has to be certified the U.S. will lose a suitable human resource that can contribute to the economy.
- **Both cases are equally important.**
- As such, we want the **F1 score to be maximized**
 - The greater the F1 score, the higher the chances of minimizing both false positives and false negatives

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Model Building - Original Data

- We start by building multiple ML models with the original data at hand.
- The models built and their performance score are provided below

	Decision Tree	Bagging	Random Forest	AdaBoost	Gradient Boosting Machine	XGBoost
Training Set F1 Score	1.000	0.989	1.000	0.819	0.829	0.895
Validation Set F1 Score	0.743	0.774	0.805	0.817	0.827	0.810

- **The Gradient Boost classifier has a good performance and best generalization**, followed by the AdaBoost classifier and XGBoost classifier respectively.

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Model Building - Oversampled Data

- Since the classes are imbalanced, we try addressing this using oversampling first.
- SMOTE (Synthetic Minority Over-sampling Technique) is used to oversample the minority class data to the same proportion as the majority class
- We then build the same models as before using the oversampled data.

	Decision Tree	Bagging	Random Forest	AdaBoost	Gradient Boosting Machine	XGBoost
Training Set F1 Score	1.0	0.987	0.999	0.798	0.807	0.870
Validation Set F1 Score	0.732	0.766	0.796	0.811	0.817	0.812

- **The Gradient Boost classifier has a good performance and best generalization**, followed by the XGBoost classifier and AdaBoost classifier respectively.

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Model Building - Undersampled Data

- We then try addressing class imbalance using undersampling first.
- Random Undersampling is used to undersample the majority class data to the same proportion as the minority class
- We then build the same models as before using the oversampled data.

	Decision Tree	Bagging	Random Forest	AdaBoost	Gradient Boosting Machine	XGBoost
Training Set F1 Score	1.0	0.980	1.0	0.705	0.728	0.872
Validation Set F1 Score	0.695	0.705	0.741	0.761	0.776	0.745

- **The Gradient Boost classifier has the best generalization**, followed by the AdaBoost classifier and XGBoost classifier respectively.

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Model Performance - Observations

- Gradient Boosting, XGBoost, and AdaBoost models demonstrate consistent and good performance along with much better generalization capabilities than the other 3 models for all 3 datasets - the original, the oversampled, and the undersampled.
- So, we will tune these three models to see if we can improve their performance.
- We will also try to tune the random forest model and see if we can solve the problem of overfitting and enhance its performance.
- We will tune the Gradient Boosting classifier and AdaBoost classifier using the original data as it gives better results for these two models.
- The XGBoost classifier and Random Forest classifier will be tuned using the oversampled data.

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Tuning Random Forest Classifier

- Grid Search with cross-validation was used for hyperparameter tuning
- The hyperparameter space checked and best hyperparameters obtained are as follows:

Hyperparameter	n_estimators	max_depth	min_samples_split
Hyperparameter Space	10, 20, 30, 40, 50, 60, 70, 80, 90, 100	2, 3, 5	3, 5, 7
Best Value Obtained	70	5	7

- The performance of the tuned Random Forest model is as follows:

	Accuracy	Precision	Recall	F1 Score
Training Set Performance	0.739	0.746	0.921	0.825
Validation Set Performance	0.739	0.748	0.917	0.824

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Tuning AdaBoost Classifier

- Grid Search with cross-validation was used for hyperparameter tuning
- The hyperparameter space checked and best hyperparameters obtained are as follows:

Hyperparameter	n_estimators	max_depth	learning_rate
Hyperparameter Space	50, 75, 100	2, 3	0.01, 0.04, 0.07, 0.1
Best Value Obtained	100	3	0.1

- The performance of the tuned AdaBoost model is as follows:

	Accuracy	Precision	Recall	F1 Score
Training Set Performance	0.749	0.776	0.876	0.823
Validation Set Performance	0.748	0.778	0.871	0.822

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Tuning Gradient Boost Classifier

- Grid Search with cross-validation was used for hyperparameter tuning
- The hyperparameter space checked and best hyperparameters obtained are as follows:

Hyperparameter	n_estimators	max_features	learning_rate	subsample
Hyperparameter Space	50, 75, 100	0.7, 0.8, 0.9, 1.0	0.01, 0.04, 0.07, 0.1	0.7, 0.9
Best Value Obtained	50	0.9	0.07	0.9

- The performance of the tuned Gradient Boost model is as follows:

	Accuracy	Precision	Recall	F1 Score
Training Set Performance	0.752	0.773	0.888	0.827
Validation Set Performance	0.753	0.776	0.886	0.827

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Tuning XGBoost Classifier

- Grid Search with cross-validation was used for hyperparameter tuning
- The hyperparameter space checked and best hyperparameters obtained are as follows:

Hyperparameter	n_estimators	learning_rate	gamma
Hyperparameter Space	50, 75, 100	0.01, 0.04, 0.07, 0.1	1, 3
Best Value Obtained	100	0.1	1

- The performance of the tuned XGBoost model is as follows:

	Accuracy	Precision	Recall	F1 Score
Training Set Performance	0.762	0.745	0.979	0.846
Validation Set Performance	0.725	0.725	0.946	0.821

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Model Performance Comparison

Training Performance

	Tuned Random Forest	Tuned AdaBoost	Tuned Gradient Boosting Machine	Tuned XGBoost
Accuracy	0.739	0.749	0.752	0.762
Precision	0.746	0.776	0.773	0.745
Recall	0.921	0.876	0.888	0.979
F1 Score	0.825	0.823	0.827	0.846

Validation Performance

	Tuned Random Forest	Tuned AdaBoost	Tuned Gradient Boosting Machine	Tuned XGBoost
Accuracy	0.739	0.748	0.753	0.725
Precision	0.748	0.778	0.776	0.725
Recall	0.917	0.871	0.886	0.946
F1 Score	0.824	0.822	0.821	0.821

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Final Model Selection

- The tuned Random Forest classifier exhibits the best performance for visa approval prediction.
- Since the Tuned Random Forest classifier gave the best and generalized performance on the test dataset, we will use it as our final model.
- The test set performance of the tuned Random Forest model is as follows:

	Accuracy	Precision	Recall	F1 Score
Test Set Performance	0.728	0.738	0.917	0.818

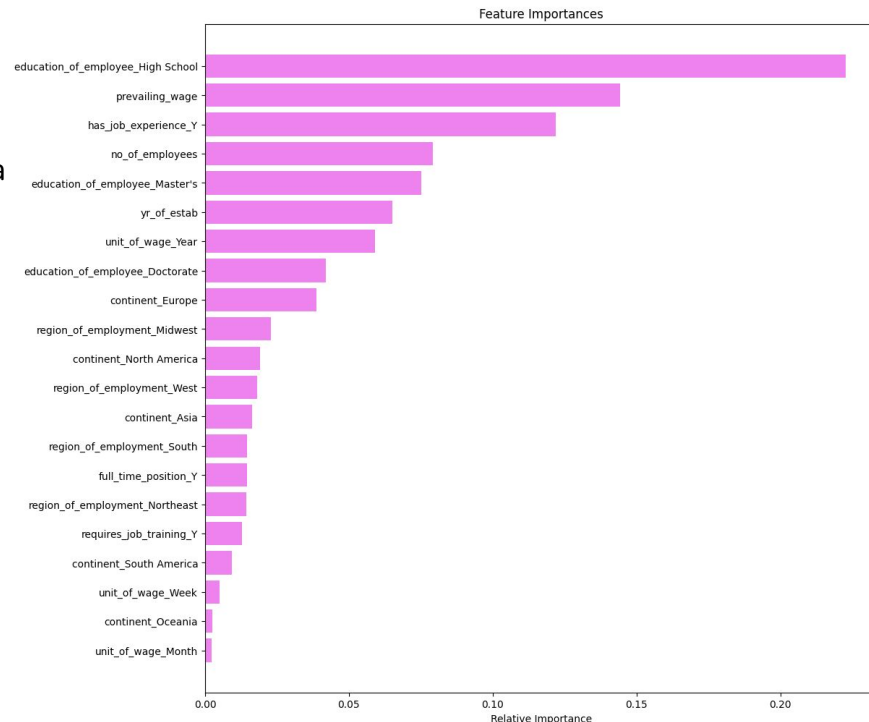
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Random Forest Classifier - Feature Importances

- The top 3 attributes affecting the tuned Random Forest model's visa approval predictions are:
 - education_of_employee_High School:** This is a categorical feature in the Random Forest model indicating the employee's education level to be high school.
 - prevailing_wage:** Average wage paid to similarly employed workers in a specific occupation in the area of intended employment.
 - has_job_experience_Y:** This variable indicates that the employee holds prior job experience.
- While other attributes may impact model performance, the above features are more likely the key determinants of visa application approval or denial.



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Actionable Insights

- **Education Level:** Applicants with at least a Bachelor's degree have a higher likelihood of visa approval. Advanced degrees such as Master's and Doctorates increase approval chances even further, especially in specific regions.
- **Job Experience:** Having job experience is a critical factor for visa certification. Applicants with no prior work experience are more likely to have their applications denied.
- **Prevailing Wage:** The median prevailing wage for certified applicants is around \$72,000, indicating that a higher wage corresponds to greater visa approval chances. In contrast, those with a prevailing wage around \$65,000 are more likely to face denial.
- **Unit of Wage:** Applications with a yearly wage unit have a higher chance of approval compared to those with hourly wages. Hourly wage applicants are more often associated with denied applications.
- **Regional Employment Trends:** Visa approval rates vary by region. The Midwest shows higher approval rates, while educational requirements differ by region:
 - High school diploma: More required in the South and Northeast.
 - Bachelor's degree: Primarily sought in the South and West.
 - Master's degree: Mostly required in the Northeast and South.
 - Doctorates: Mainly in demand in the West and Northeast.

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Recommendations

- **Target Higher Education Profiles:** Focus on applicants with at least a Bachelor's degree, and prioritize Master's or Doctorate holders, particularly for applications in regions like the Northeast and West where advanced degrees are in higher demand.
- **Encourage Competitive Wage Offers:** Employers should aim to offer salaries at or above the median prevailing wage of \$72,000 to enhance visa approval chances. Conducting market research to align wages with industry standards may also help improve certification rates.
- **Prioritize Yearly Wage Unit:** Companies should consider offering yearly wage packages rather than hourly wages to increase the likelihood of visa approval. A shift toward yearly wages aligns with successful applicant profiles.
- **Leverage Regional Employment Needs:** Tailor applicant recruitment based on regional demand. For example, focus on high school graduates for Southern regions and Master's degree holders for the Northeast. This targeted approach could improve visa success rates by addressing specific regional talent shortages.
- **Gather More Detailed Applicant Information:** Employers should collect and analyze additional details about candidates, such as their field of specialization and years of experience. This data can provide deeper insights and help in making better hiring decisions, leading to higher chances of visa approval.

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APPENDIX

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Data Overview

- The dataset has **25480 rows** and **12 columns**
- The dataset does not have any null or duplicate values.
- The datatypes of the different columns of the dataset are given below:

Data Type	No. of variables
object	9
float64	1
int64	2

- The case_id contains unique identifiers for each row. This column is dropped from the data as it is not useful for any model building or EDA processes.

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Data Overview

- The statistical description of the numerical columns of the dataset is given below.

	count	mean	std	min	25%	50%	75%	max
no_of_employees	25480	5667.04	22877.92	-26	1022	2109	3504	602069
yr_of_estab	25480	1979.4	42.36	1800	1976	1997	2005	2016
prevailing_wage	25480	74455.81	52815.94	2.136	34015.48	70308.21	107735.51	319210.27

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Data Overview

From the descriptive statistics, we can conclude that:

- The range of the number of employees in a company is huge. There are some anomalies in the data as we can see that the minimum number of employees is equal to -26, which is not possible. We will have to fix this.
- The year of establishment of companies ranges from 1800 to 2016, which seems fine.
- The average prevailing wage is 74455.81. There's also a very huge difference in the 75th percentile and maximum value which indicates there might be outliers present in this column.

We fix the negative values in the no_of_employees column by taking their absolute values.

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Power Ahead!

